



A conditionally heteroskedastic model with time-varying coefficients for daily gas spot prices[☆]

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ABSTRACT

This paper examines the relationship between gas spot prices at the Zeebrugge market, one-month ahead Brent prices and temperatures over the period 2000–2005. A cointegration analysis is carried out and it is discovered that a cointegration relationship exists between the three series. To take into account the influence of temperature on the gas volatility, a GARCH(1,1) model with temperature-dependent coefficients is considered. Stability and estimation properties are discussed. An empirical finding is the existence of distinct volatility regimes for the volatility of gas prices, depending on the temperature level.

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1. Introduction

Following the deregulation of natural gas markets in Europe, natural gas transactions between producing countries and retailers – historically run by long term contracts indexed on crude oil – were diversified through new financial markets (National Balancing Point in the UK, Zeebrugge market in Belgium), where it could be freely sold at different time horizons. This restructuring has generated uncertainty, requiring the development of appropriate valuation and risk management strategies. Such strategies require an appropriate modeling of the price volatility. The standard GARCH models of Engle (1982) and Bollerslev (1986), which arguably constitute the most important class of models for financial data, may be inadequate for energy prices. The reason is that energy prices are subject to pronounced daily seasonal patterns, which may not only concern the conditional mean but also the volatility.

The contribution of this article is twofold. First, we employ Johansen's (1988, 1995) methodology to explore the joint dynamics of gas prices at the Zeebrugge market, one-month ahead Brent prices and temperatures. More precisely, we examine the existence of a

long-term relationship between those series, in order to model their joint dynamics via a vector error-correction model. Second, we employ for the gas prices volatility a class of models recently introduced by Regnard and Zakoïan (2010). The aim is to capture the seasonal patterns of the volatility induced by temperature. In this model, the parameters associated with the volatility dynamics are indexed by the temperature level. Similar dynamics have been proposed for the conditional mean by Azrak and Mélard (2006), Bibi and Francq (2003), and Francq and Gautier (2004a, 2004b). In such models, the coefficients depend on the realizations of an exogenous process which, in the present paper, is related to the temperature levels. Having removed, in the first part of the study, the linear influence of temperature, we include temperature levels, in a second step, as exogenous variables for the gas prices volatility.

Previous works have introduced seasonal behaviors in volatility models. The periodic ARCH model, which the model of this paper generalizes, was introduced by Bollerslev and Ghysels (1996) and studied by Aknouche and Bibi (2009). It is able to capture those seasonal behaviors in the conditional variance. However, in this model the different regimes appear in a purely periodic succession and it may be worth introducing more flexibility.

A GARCH model with regression effects and scaled by seasonal factors has been recently proposed for electricity prices by Koopman, Ooms and Carnero (2007).

The article is organized as follows. Section 2 provides information about the data set and derives stationarity tests. A cointegration analysis is proposed in Section 3, through the estimation of a vector error correction model, involving daily gas prices, Brent prices and

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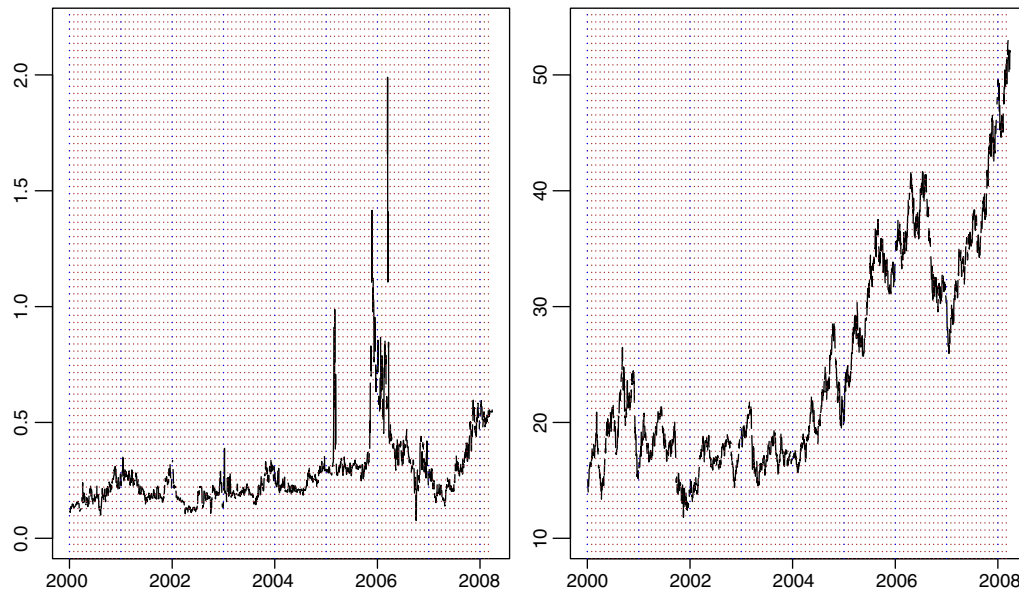


Fig. 1. Daily series of gas prices G_t (left panel) and Brent prices B_t (right panel).

temperatures. Section 4 introduces a volatility model with temperature-dependent coefficients and reviews its main probability properties.

The properties of the QML (Quasi-Maximum Likelihood) estimator are discussed and a consistent estimator is derived for the asymptotic covariance matrix. In Section 5, the volatility of gas prices is studied. The proposed model is fitted with up to seven volatility regimes depending on the temperature. The different specifications are tested, and compared via out-of-sample predictions. Asymmetric extensions of the model are also considered. Section 6 concludes. A technical proof is given in the appendix.

2. Data and stationarity tests

We have a sample of daily gas spot prices (G_t) at the Zeebrugge market, one-month ahead Brent prices (B_t) and temperatures (T_t), from January, 4, 2000 to December, 21, 2005 ($n=1,272$ cotation dates, excluding week-ends).¹ Fig. 1 displays a plot of the gas and Brent series, (G_t) and (B_t) on the whole period. Let $g_t = \log G_t$ and $b_t = \log B_t$ denote the gas and Brent prices in logarithms. The three series (g_t), (b_t) and (T_t) are displayed in Fig. 2. The graphs of the prices seem to indicate departures from stationarity, while the temperatures series shows a seasonal behavior.

To test the non stationarity, or the stationarity, of the series of prices we considered three tests: the augmented Dickey–Fuller (ADF) test introduced by Said and Dickey (1984), the KPSS test developed by Kwiatkowski, Phillips, Schmidt and Shin (1992), and the ERS test proposed by Elliott, Rothenberg and Stock (1996). The ADF test examines the null hypothesis that a time series is first-order integrated ($I(1)$) against the alternative that it is stationary, assuming that the dynamics in the data have an ARMA structure. On the other hand, the KPSS test is the most commonly used stationarity test (under the null hypothesis, the series is stationary). Finally, ERS tests, referred to as efficient unit root tests, can have substantially higher power than the ADF unit root tests especially when the autoregressive parameter is close to unity. These statistics are computed for the series g_t and b_t , with a constant or a deterministic trend included in the

regression. The results displayed in Table 1 confirm that the two series are $I(1)$. Similar tests conducted for the series T_t lead to the same conclusion.

Now we proceed to investigate the existence of a cointegrating relationship between the three series.

3. Cointegration analysis

To capture the joint behavior of the series of gas, Brent prices and temperatures, we consider a vector error correction model (VECM). There is a growing literature examining the cointegration relationships between different energy prices. Asche, Osmundsen and Sandmark (2006) discuss the cointegration between UK natural gas, Brent oil and electricity prices before and after the opening of the Interconnector in 1998. Bachmeier and Griffin (2006) found evidence of cointegration between crude oil, natural gas and coal in the USA. Panagiotidis and Rutledge (2007) found evidence of a cointegration relationship between the UK wholesale gas prices and the Brent over the period 1996–2003, contradicting the assumption that gas prices and oil prices are decoupled since the liberalization of gas markets in Europe. A distinctive feature of our analysis is to include not only energy prices but also temperature in our multivariate time series. The reason is that, in a second step, we will treat temperature as an exogenous variable explaining the price volatilities. It is thus natural to introduce it also in the preliminary analysis which aims at filtering from linear effects.

Recall that, in Johansen's (1988, 1995) notation, a p -dimensional VECM takes the form

$$\Delta y_t = \sum_{i=1}^{k-1} \Gamma \Delta y_{t-i} + \Pi y_{t-1} + \mu + u_t$$

where Δ is the first difference operator, y_t is a $p \times 1$ vector of $I(1)$ variables, μ is a drift parameter, (u_t) is a white noise, $\Pi = \alpha\beta'$ is a $p \times p$ matrix where α and β are $p \times r$ full-rank matrices, with β containing the r cointegrating vectors and α carrying the loadings in each of the r vectors. Visual inspection of the graphs suggests that oil prices have an impact on gas prices with a delay of roughly 13 weeks. Table 2, showing the residual variance in the linear regression of g_t on $b_{t-\tau}$ and T_t , confirm that $\tau=91$ days is an appropriate choice for the delay between the two energy prices (a longer delay, $\tau=140$, could be considered as well but the variance reduction is small compared to

¹ These data are purchasable from APX site for the Zeebrugge average day ahead spot prices and Brent one month ahead at the address: <http://www.apxindex.com/> and from the metoffice for the UK mean temperature at the address: <http://www.metoffice.gov.uk>. Precise description of these data is also available on this sites.

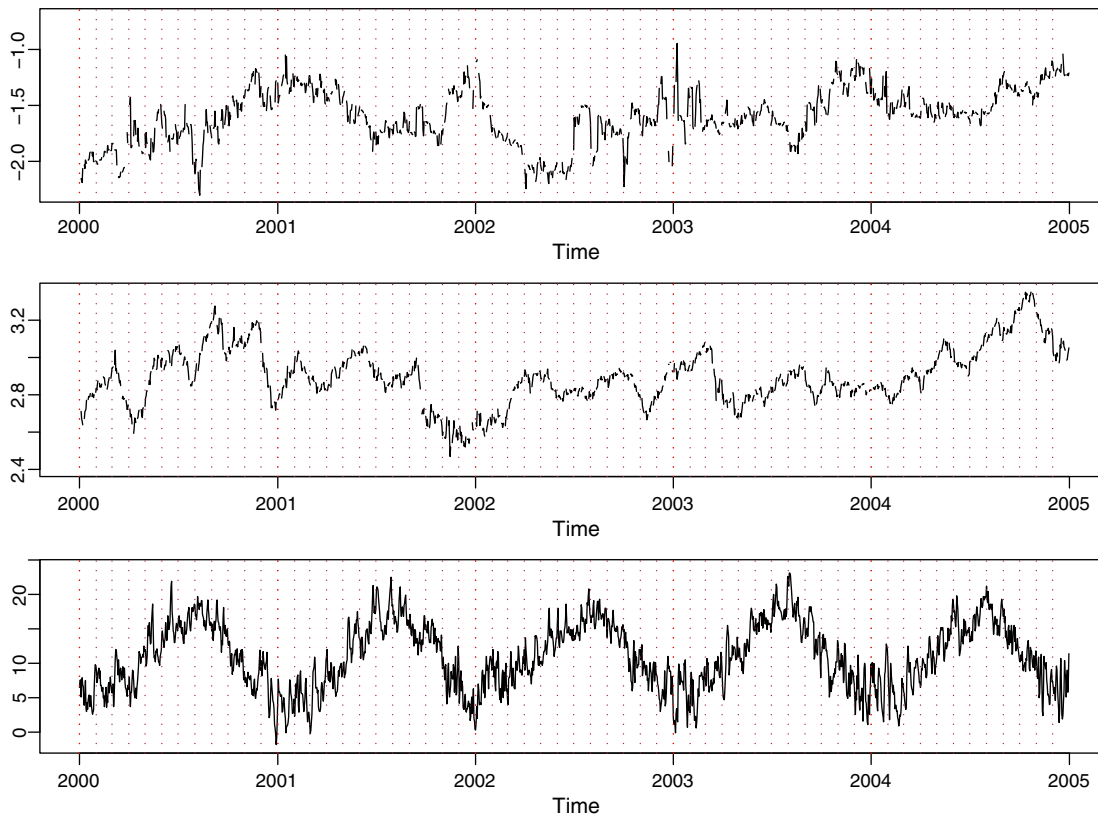


Fig. 2. Daily series of gas log-prices g_t (upper panel), Brent log-prices b_t (middle panel) and temperatures T_t (lower panel).

$\tau = 91$). Let then $y_t = (g_t, b_{t-\tau}, T_t)$ where $\tau = 91$ days. As can be seen from Table 3, the Johansen trace test rejects the null hypothesis of zero cointegrating vectors between the components of y_t . The existence of $r = 1$ cointegrating relation is not rejected and the estimated cointegration vector is, by renormalizing so that the first element be in unity, $\hat{\beta} = (1, -1.0809, 0.0194)$. Let $c_t = g_t - 1.080b_{t-\tau} + 0.019T_t + 4.46$.

The estimated VECM is as follows. For ease of presentation, insignificant coefficients, at the 5% level, have been omitted. The standard errors appear in parenthesis.

$$\Delta g_t = \underset{(0.012)}{-0.077c_t} + \underset{(0.030)}{0.056\Delta g_{t-1}} - \underset{(0.001)}{0.010\Delta T_{t-4}} - \underset{(0.029)}{0.103\Delta g_{t-5}} \\ - \underset{(0.029)}{0.091\Delta g_{t-6}} - \underset{(0.028)}{0.087\Delta g_{t-8}} - \underset{(0.001)}{0.003\Delta T_{t-8}} + \epsilon_t$$

$$\Delta b_{t-\tau} = \zeta_t$$

$$\Delta T_t = \underset{(0.030)}{-0.218\Delta T_{t-1}} - \underset{(0.031)}{0.280\Delta T_{t-2}} - \underset{(0.032)}{0.225\Delta T_{t-3}} - \underset{(0.032)}{0.207\Delta T_{t-4}} \\ - \underset{(0.032)}{0.135\Delta T_{t-5}} - \underset{(0.032)}{0.107\Delta T_{t-6}} - \underset{(0.030)}{0.067\Delta T_{t-8}} + \xi_t.$$

It is worth noting that for the Brent prices, no significant linear influence of the past variables is detected. Results not reported here show that the process $(\epsilon_t, \zeta_t, \xi_t)$ passes the diagnostic tests for the absence of autocorrelation. Other devices, such as the seasonal cointegration introduced by [Hylleberg, Engle, Granger and Yoo \(1990\)](#), could be considered to filter seasonality of the temperature series. This is left for further investigations.

Having modeled the conditional mean of the commodity prices and the temperature through a VECM model, we investigate the presence of volatility effects, that is a non constant conditional variance, in the noise. A standard approach would be to use a multivariate GARCH to capture conditional heteroskedasticity in the vector $(\epsilon_t, \zeta_t, \xi_t)$. Results presented in the appendix for an asymmetric Constant Conditional Correlation (CCC) model show that the volatilities of the three series are mainly influenced by their own past values. Coefficients corresponding to cross-effects are very small, if not null, except in the correlation matrix. However, this approach fails to capture seasonal patterns in the price volatilities induced by temperature. A natural extension would be to consider a periodic multivariate GARCH model, as the one studied by [Osborn, Savva and Gill \(2008\)](#). Because our aim is to capture seasonal though not exactly periodic patterns, we follow a different approach in the next section.

Table 1

Unit root tests statistics for the series b_t , g_t , Δb_t and Δg_t with constant (“-c”) or trend (“-t”).

	b_t		g_t		Δb_t		Δg_t	
	Value	p-val	Value	p-val	Value	p-val	Value	p-val
ADF-c	-2.79	0.25	-3.94	0.01	-1.359	<0.01	-4.13	<0.01
ADF-t	-2.79	0.75	-3.94	0.99	-1.359	<0.01	-4.18	<0.01
KPSS-c	15.06	0.01	8.75	0.01	0.0153	<0.01	0.0328	<0.01
KPSS-t	2.54	0.01	0.54	0.01	0.0154	<0.01	0.0338	<0.01
ERS-c	0.34	>0.05	-1.93	>0.05	-3.6471	<0.01	-3.923	<0.01
ERS-t	-2.50	>0.05	-5.20	<0.01	-7.0062	<0.01	-7.309	<0.01

Table 2Variances of residuals from the linear regression (with constant) of g_t on $(b_t - \tau, T_t)$.

τ	0	14	35	91	140	210
Variance of residuals ($\times 100$)	5.72	5.49	5.39	4.13	3.79	4.92

For the rest of the paper, we focus on the noise of the gas price series (ϵ_t) .

4. A volatility model driven by temperature for the gas prices residuals

Fig. 3 displays the series of residuals ϵ_t for gas prices. The empirical autocorrelation function (ACF) and partial autocorrelation function (PACF) of (ϵ_t) are displayed in Fig. 4. The standard significance bands, $\pm 1.96/\sqrt{n}$, displayed in dotted lines, are asymptotically valid for independent white noises. To allow for possible nonlinearities, we considered the asymptotic bands derived by Francq and Zakoian (2009). These bands are based on a correction of the standard Bartlett formula and are asymptotically valid, for the ACF and PACF, under mild regularity conditions except the existence of fourth-order moments. The bands in the left panels are obtained under the assumption of a GARCH(1,1) white noise. The bands in the right panels do not rely on any parametric model, and are valid for a weak white noise, that is a sequence of centered and uncorrelated variables.² From these figures, it is clear that this series has the characteristics of a white noise. However, white noise meaning uncorrelatedness, it does not preclude the existence of dependencies between ϵ_t and its past values. Such dependencies are usually depicted by inspecting the autocorrelograms of the series (ϵ_t^2) . The ACF and PACF displayed in Fig. 5 clearly show that (ϵ_t^2) is an autocorrelated sequence. Significant autocorrelations of squares may indicate the presence of conditional heteroscedasticity. It is thus natural to consider GARCH models for such squared residuals.

Recall that the GARCH(1,1) model of Engle (1982) and Bollerslev (1986) is defined by

$$\epsilon_t = \sigma_t \eta_t, \quad \sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad t \in \mathbb{Z}, \quad (1)$$

where $\omega > 0, \alpha, \beta \geq 0$ and (η_t) is a sequence of independent and identically distributed (iid), centered variables with unit variance. The GARCH formulation has proven very successful to model financial series but for energy prices the existence of fixed coefficients ω, α and β may constitute a serious limitation. Indeed, as we have seen from the VECM analysis, the temperature has an important influence on the conditional mean of the gas prices and it seems desirable to allow a similar influence on the conditional variance.

To show the influence of temperature on the volatility of (ϵ_t) , we draw in Fig. 6 the ACF of (ϵ_t^2) corresponding to different temperature levels (which will be discussed later on), namely, for $i = 1, 2, 3$,

$$\hat{\rho}^{(i)}(h) = \frac{\sum_{t=1}^n \left(\left\{ \epsilon_t^2 - \hat{m}_i \right\} 1_{\{T_t \in C_i\}} \right) \left(\epsilon_{t-h}^2 - \hat{m} \right)}{\left\{ \sum_{t=1}^n \left(\epsilon_t^2 - \hat{m}_i \right)^2 1_{\{T_t \in C_i\}} \right\}^{1/2} \left\{ \sum_{t=1}^n \left(\epsilon_{t-h}^2 - \hat{m} \right)^2 \right\}^{1/2}} \quad (2)$$

$$\hat{m}_i = \frac{\sum_{t=1}^n \epsilon_t^2 1_{\{T_t \in C_i\}}}{\sum_{t=1}^n 1_{\{T_t \in C_i\}}}, \quad \hat{m} = \frac{\sum_{t=1}^n \epsilon_t^2}{n},$$

$$C_1 = \{T < 9\}, \quad C_2 = \{T \in [9, 14]\}, \quad C_3 = \{T > 14\}.$$

Squared observations corresponding to the lowest temperatures appear to be more autocorrelated than squared observations of the

Table 3

The trace test of the cointegration rank.

H_0 : rank $(\leq)$	Trace test	Critical values		
		10%	5%	1%
2	6.10	7.52	9.24	12.97
1	18.43	17.85	19.96	24.60
0	48.25	32.00	34.91	41.07

two other regimes. It seems therefore natural to introduce temperature in the volatility specification.

4.1. Model specification

Starting from Model (1), a natural approach for taking into account the influence of the temperature is to make the coefficients ω, α and β depend on the level of temperature at time t , denoted s_t . For instance, if two levels of temperature are considered, say “low” and “high”, s_t will take two values (by convention 1 and 2) and each of the coefficients ω, α and β will take two values. More generally, the model we consider in this paper was introduced by Regnard and Zakoian (2010) and is given by

$$\epsilon_t = \sigma_t \eta_t, \quad \sigma_t^2 = \omega(s_t) + \alpha(s_t) \epsilon_{t-1}^2 + \beta(s_t) \sigma_{t-1}^2, \quad t \in \mathbb{Z} \quad (3)$$

where (η_t) is as in Model (1) the temperature level s_t is valued in a finite set $E = \{e_1, \dots, e_d\}$; the functions $\omega(\cdot), \alpha(\cdot), \beta(\cdot)$ are defined on E with values in $\mathbb{R}^+$ with $\omega(\cdot) > 0$.

For each level of temperature, the volatility is that of a standard GARCH(1,1) model. Thus, if this level remains constant in some period, the volatility is governed by a standard GARCH. When another level of temperature is reached, the specification of the volatility changes. Other variables could be considered for the regime variable. However, the temperature can be considered as exogenous because its dynamics is clearly not influenced by the gas prices. Introducing other commodity prices as exogenous variables would be more disputable from an economic point of view.

This model has connections with several classes of models introduced in the econometric and statistical literatures, which postulate the existence of different regimes. A detailed account on mixture models can be found in Frühwirth-Schnatter (2006), and a mixture bilinear model was recently proposed by Aknouche and Rabehi (2010). Markov-switching models (MSM) were introduced by Hamilton (1989) in the context of ARMA models, and by Hamilton and Susmel (1994) in the context of GARCH. Susmel and Thompson (1997) proposed a MSM for the volatility of the natural gas price change series. Our model and MSM have a completely different interpretation. In the latter models, the regime-change mechanism is governed by a *non observable* (or hidden) variable. In our specification, it is governed by an *observable* process. In this sense, our model is also related to the class of Threshold GARCH models (see Zakoian (1994), Li and Li (1996)) but with the difference that the regime shifts in the conditional variance are exogenous, instead of being governed by a past endogenous variable in Threshold GARCH models.

4.2. Stability properties

The probabilistic properties of Model (3) have been established by Regnard and Zakoian (2010) (hereafter RZ). Assume

A0. (s_t) is a realization of a process (S_t) which is stationary, ergodic, defined on the same probability space $(\Omega, \mathcal{A}, \mathbb{P})$ as (η_t) , and independent of (η_t) ,

and introduce

$$\pi_j = P(S_t = e_j), \quad j = 1, \dots, d \quad \text{and} \quad a(x, y) = \alpha(x)y^2 + \beta(x).$$

² We used the R-codes available from the web site <http://perso.univ-lille3.fr/francq/Christian-Francq-Generalized-Bartlett-Formula.html>.

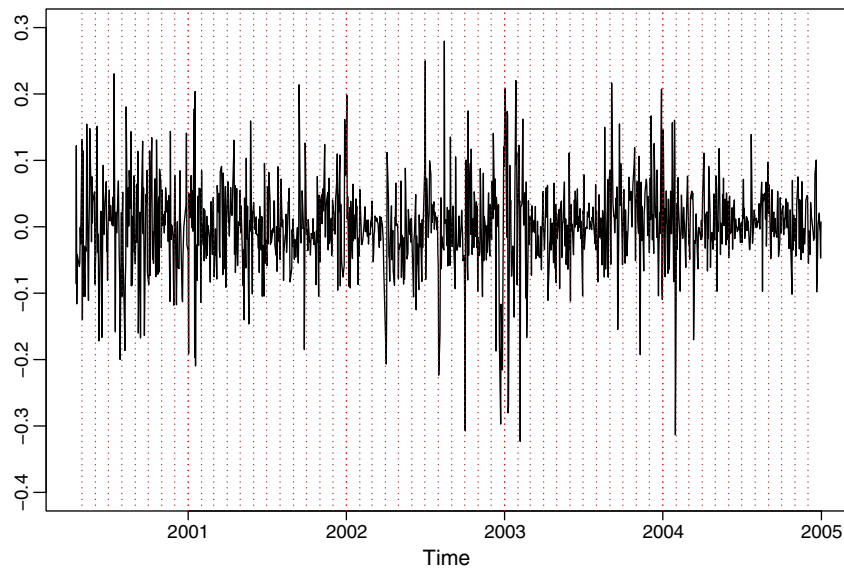


Fig. 3. Series (ϵ_t) for the gas prices.

Then, if

$$\gamma_0 := \sum_{j=1}^d \pi_j E\{\log a(e_j, \eta_0)\} < 0, \quad (4)$$

Model (3) admits a nonanticipative nonexplosive solution (ϵ_t) . When $\gamma_0 > 0$, the process is explosive: for any initial value σ_0^2 , we have $\sigma_t^2 \rightarrow +\infty$, a.s. $t \rightarrow \infty$. Condition (4) can thus be referred to as a *stability* condition, not a stationarity condition since the solution is not a stationary process when $d > 1$.

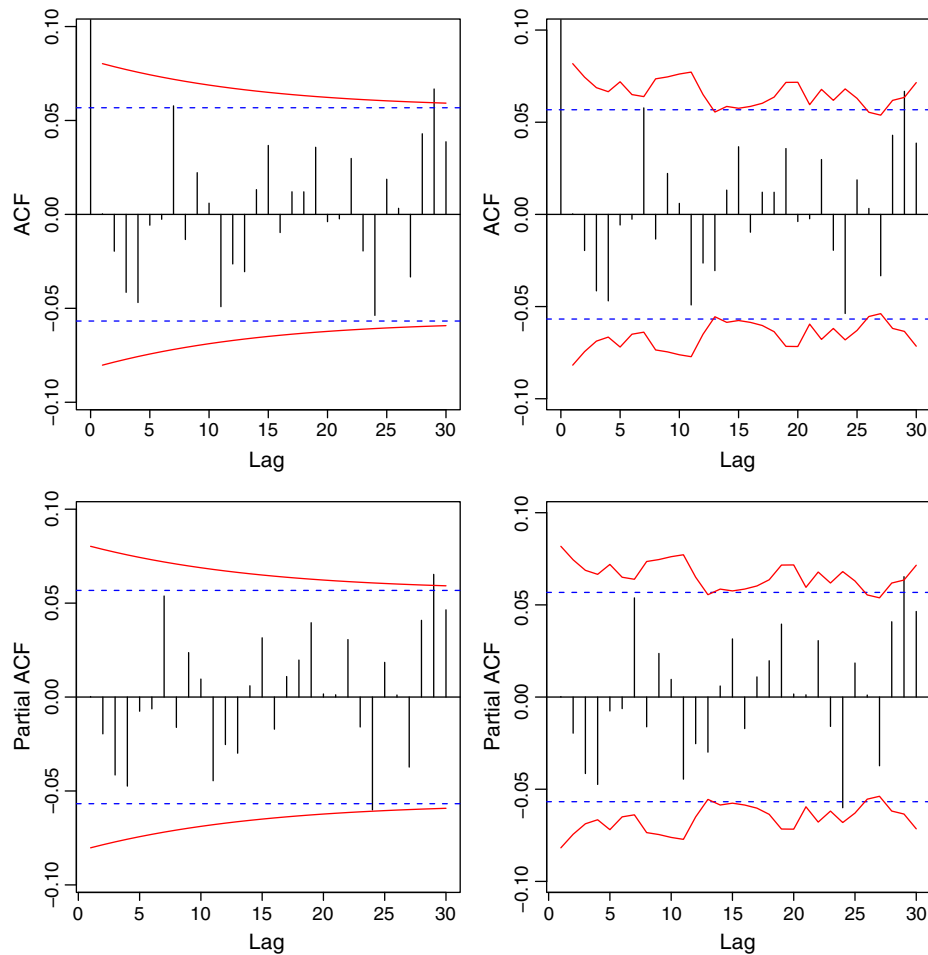


Fig. 4. Empirical ACF and PACF of the series (ϵ_t) and significance bands at the 95% level. The bands $\pm 1.96 / \sqrt{n}$, for independent white noises, are displayed in dotted lines. The bands in the left panels are obtained under the assumption of a GARCH(1,1) white noise. Nonparametric bands for weak white noises are displayed in the right panels.

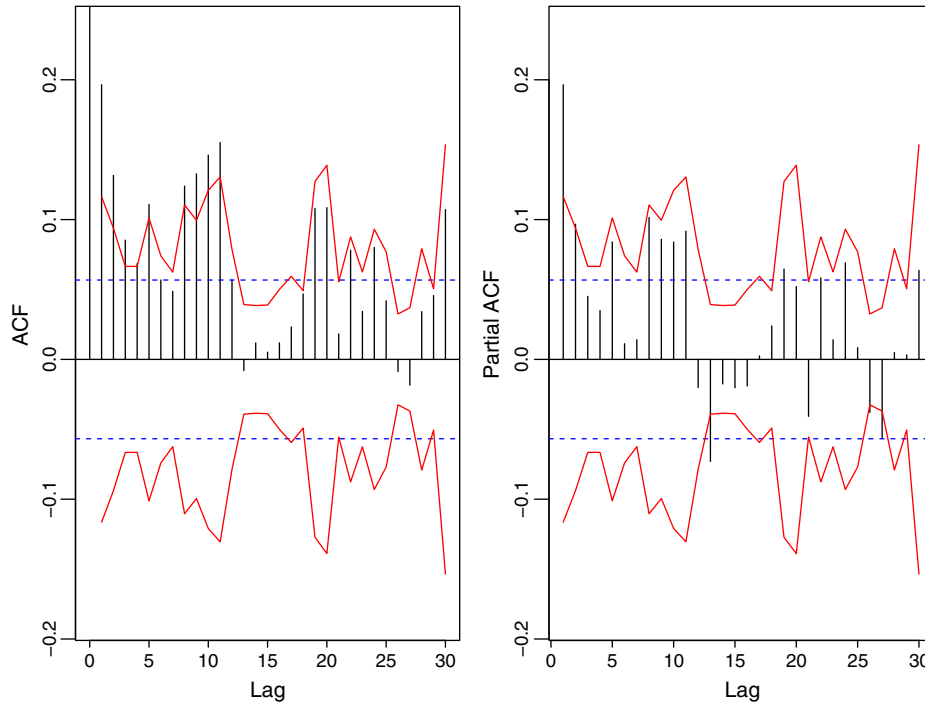


Fig. 5. Empirical ACF and PACF of the series (ϵ_t^2) and significance bands at the 95% level. The bands $\pm 1.96 / \sqrt{n}$ are displayed in dotted lines. Nonparametric significance bands are displayed in full lines.

In the ARCH(1) case (no coefficients β), Condition (4) takes the more explicit form $\prod_{j=1}^d \alpha^{\eta_j}(j) < e^{-E \log \eta_0^2}$. It can also be noted that the stability of the GARCH(1,1) in each regime, that is

$$E\{\log \alpha(j)\eta_0^2 + \beta(j)\} < 0, \quad j = 1, \dots, d$$

is sufficient (but not necessary) for the global stability. A necessary condition for Eq. (4) is given by $\prod_{j=1}^d \beta^{\eta_j}(j) < 1$.

Existence of moments requires stronger conditions. Letting $\mu(x) = \alpha(x) + \beta(x)$, $\lambda(x) = \kappa_{\eta} \alpha(x)^2 + 2\alpha(x)\beta(x) + \beta(x)^2$, where $\kappa_{\eta} = E\eta^4$, we have

$$\prod_{j=1}^d \mu(e_j)^{\eta_j} < 1 \Rightarrow E\epsilon_t^2 < \infty, \quad \text{and} \quad \prod_{j=1}^d \lambda(e_j)^{\eta_j} < 1 \Rightarrow E\epsilon_t^4 < \infty. \quad (5)$$

Similar conditions hold for the existence of higher-order moments. It is important to note that, when existing, those moments are time-dependent (except in the case $d=1$ which corresponds to the standard GARCH(1,1) model).

4.3. QML estimation

The consistency and asymptotic normality of the QMLE have been proven under mild conditions by Berkes, Horváth and Kokoszka (2003) and Francq and Zakoian (2004) for standard GARCH and ARMA-GARCH models. RZ showed that these properties can be extended to the model of this paper.

Let θ denote the vector of parameters,

$$\theta = (\omega(e_1), \dots, \omega(e_d), \alpha(e_1), \dots, \alpha(e_d), \beta(e_1), \dots, \beta(e_d))',$$

with true value θ_0 . The parameter is assumed to belong to a parameter space $\Theta \subset]0, +\infty[^d \times]0, \infty[^{2d}$. The sequence (s_t) is observed, and d is known a priori. Let $(\epsilon_1, \dots, \epsilon_n)$ be a realization of length n of the nonanticipative solution (ϵ_t) . A QMLE of the true parameter value, θ_0 , is defined as any measurable solution $\hat{\theta}_n$ of

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{t=1}^n \frac{\epsilon_t^2}{\hat{\sigma}_t^2} + \log \hat{\sigma}_t^2 \quad (6)$$

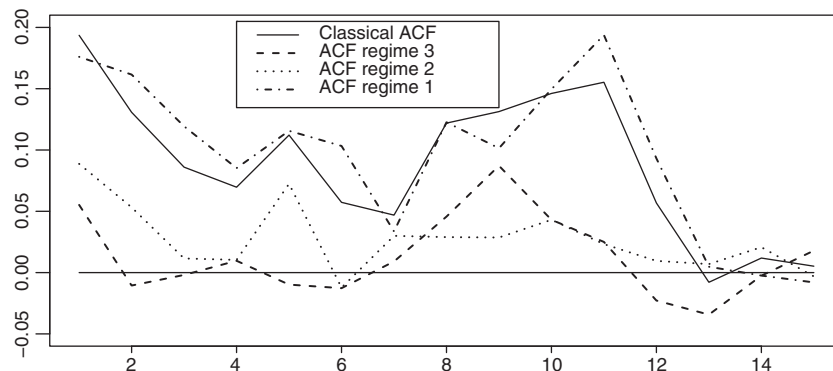


Fig. 6. Empirical ACF of the series (ϵ_t^2) taking into account the temperature levels. The ACF for the regime $i=1,2,3$ are defined in Eq. (2).

where the $\tilde{\sigma}_t^2$ are defined recursively, for $t \geq 2$, by

$$\tilde{\sigma}_t^2 = \tilde{\sigma}_t^2(\theta) = \omega(s_t) + \alpha(s_t)\epsilon_{t-1}^2 + \beta(s_t)\tilde{\sigma}_{t-1}^2,$$

with initial values ϵ_0^2 and $\tilde{\sigma}_0^2$. RZ showed that, under appropriate assumptions,

- (1) Almost surely $\hat{\theta}_n \rightarrow \theta_0$, as $n \rightarrow \infty$,
- (2) $\sqrt{n}(\hat{\theta}_n - \theta)$ is asymptotically $\mathcal{N}(0, (\kappa_\eta - 1)J^{-1})$ distributed, where

$$J := E_{S,\eta} \left(\frac{1}{\sigma_{S,t}^4(\theta_0)} \frac{\partial \sigma_{S,t}^2(\theta_0)}{\partial \theta} \frac{\partial \sigma_{S,t}^2(\theta_0)}{\partial \theta'} \right) \quad (7)$$

is a positive-definite matrix, and $(\sigma_{S,t}^2(\theta_0))$ is the process obtained by replacing s_t by S_t in the second equation of Eq. (3).

The following examples illustrate the influence of the distributions of (S_t) and (η_t) on the asymptotic covariance matrix of the QMLE, for a 2-regime ARCH(1) model given by

$$\epsilon_t = \begin{cases} (1 + 0.1\epsilon_{t-1}^2)^{1/2} \eta_t & \text{if } s_t = 1 \\ (3 + 0.1\epsilon_{t-1}^2)^{1/2} \eta_t & \text{if } s_t = 2 \end{cases}. \quad (8)$$

Suppose that (S_t) is a Markov chain with transition probabilities $p(i,j)$. Then, if

- $p(1,1) = p(2,2) = 0.5$; $\eta_t \sim \mathcal{N}(0,1)$:

$$\text{Var}_{\text{as}}(\sqrt{n}(\hat{\theta}_n - \theta)) = \begin{pmatrix} 7.41 & 0 & -1.62 & 0 \\ 0 & 56.78 & 0 & -8.96 \\ -1.62 & 0 & 1.30 & 0 \\ 0 & -8.96 & 0 & 5.28 \end{pmatrix};$$

- $p(1,1) = p(2,2) = 0.95$; $\eta_t \sim \mathcal{N}(0,1)$:

$$\text{Var}_{\text{as}}(\sqrt{n}(\hat{\theta}_n - \theta)) = \begin{pmatrix} 3.83 & 0 & -1.33 & 0 \\ 0 & 300.51 & 0 & -53.24 \\ -1.33 & 0 & 1.58 & 0 \\ 0 & -53.24 & 0 & 32.39 \end{pmatrix};$$

- $p(1,1) = p(2,2) = 0.95$, η_t is distributed as a mixture of Gaussian distributions (with $\kappa_\eta \approx 9$):

$$\text{Var}_{\text{as}}(\sqrt{n}(\hat{\theta}_n - \theta)) = \begin{pmatrix} 11.39 & 0 & -1.92 & 0 \\ 0 & 918.26 & 0 & -77.02 \\ -1.92 & 0 & 4.21 & 0 \\ 0 & -77.02 & 0 & 87.99 \end{pmatrix}.$$

The expectation in Eq. (7) has been obtained by simulation. The presence of asymptotic covariances equal to zero for certain parameters is due to the absence of coefficient β in Model (8).

4.4. Consistent estimation of the asymptotic variance of the QMLE

To build tests and confidence intervals for the parameters of Model (3), it is essential to have a consistent estimator of the asymptotic covariance matrix of the QMLE. In view of Eq. (7), this matrix depends on the distribution of (S_t) which is unknown. However, the following result provides a consistent estimator which can be easily computed.

Proposition 1. Under the assumptions required for the asymptotic normality of $\hat{\theta}_n$ (A0-A7 in RZ), a strongly consistent estimator of J is given by

$$\hat{J}_n = \frac{1}{n} \sum_{t=1}^n \frac{1}{\tilde{\sigma}_t^4(\hat{\theta}_n)} \frac{\partial \tilde{\sigma}_t^2(\hat{\theta}_n)}{\partial \theta} \frac{\partial \tilde{\sigma}_t^2(\hat{\theta}_n)}{\partial \theta'},$$

and a strongly consistent estimator of $(\kappa_\eta - 1)J^{-1}$ is

$$(\hat{\kappa}_\eta - 1)\hat{J}_n^{-1}, \quad \text{where} \quad \hat{\kappa}_\eta = \frac{1}{n} \sum_{t=1}^n \frac{\epsilon_t^4}{\tilde{\sigma}_t^4(\hat{\theta}_n)}.$$

Proof. See appendix.

5. Empirical results for the gas volatility

The volatility models for the series ϵ_t were estimated following a specific-to-general approach, over the period April 2000 to December 2004, involving 1192 observations. To have a benchmark, the following standard one-regime GARCH(1,1) model was fitted, with standard deviations in brackets.

$$\sigma_t^2 = 0.0004 + \frac{0.17}{(0.0001)} \epsilon_{t-1}^2 + \frac{0.75}{(0.06)} \sigma_{t-1}^2 \quad (9)$$

The GARCH coefficients are close to those generally obtained for financial series, with a strong persistence in volatility ($\alpha + \beta = 0.92$).

Next, we turn to multi-regime GARCH(1,1) models, where the regimes are determined by levels of temperature. Our strategy for the regime choices is to start with a three-regime model, where the three classes of temperatures correspond roughly to one third of the total set of observations. Then, the regimes of low and high temperatures will be split in two regimes each, of approximately the same number of observations. Another five-regime model will be considered by taking into account extreme temperatures. The next model allows for a thinner decomposition with 7 regimes. Finally, a purely periodic model corresponding to the four seasons will be estimated.

The choice of three regimes with (roughly) equal number of observations leads to set $s_t = 1$ when $T_t < 9$, $s_t = 2$ when $T_t \in [9, 14]$, and $s_t = 3$ when $T_t > 14$, with frequencies in the sample

$$\hat{\pi}_1 = 0.35, \quad \hat{\pi}_2 = 0.32, \quad \hat{\pi}_3 = 0.33. \quad (10)$$

The fitted three-regime GARCH(1,1) model is as follows.

$$\sigma_t^2 = \begin{cases} 0.0003 + \frac{0.13}{(0.0002)} \epsilon_{t-1}^2 + \frac{0.80}{(0.06)} \sigma_{t-1}^2 & \text{when } T_t < 9, \\ 0.0010 + \frac{0.36}{(0.0003)} \epsilon_{t-1}^2 + \frac{0.38}{(0.16)} \sigma_{t-1}^2 & \text{when } 9 \leq T_t \leq 14, \\ 0.0004 + \frac{0.14}{(0.0002)} \epsilon_{t-1}^2 + \frac{0.76}{(0.10)} \sigma_{t-1}^2 & \text{when } T_t > 14. \end{cases} \quad (11)$$

All coefficients, except the intercept in the first regime, are significant at the 5% level. The most striking point is the difference between the volatility dynamics in the middle regime, compared to the volatilities of the two extreme regimes. The volatility of the second regime is less persistent ($\alpha(2) + \beta(2) = 0.73$) with a more convex “news-impact curve”. The impact of recent observations on the volatility is stronger than in the low- and high-temperature regimes. It can be noted that the three GARCH(1,1) models are second-order stationary, which entails the global stability with a finite time-dependent variance for ϵ_t . Note also that the marginal variances within each regime $(\omega(j)/(1 - \alpha(j) - \beta(j)))$ are roughly the same (around 0.04).

The next model is based on a decomposition of the lower and upper regimes in Eq. (11). Letting $s_t = 1$ when $T_t < 6$, $s_t = 2$ when $T_t \in [6, 9]$, $s_t = 3$ when $T_t \in [9, 14]$, $s_t = 4$ when $T_t \in [14, 16]$, and $s_t = 5$ when $T_t > 16$, the regime frequencies are given by $\hat{\pi}_1 = 0.16$, $\hat{\pi}_2 = 0.19$, $\hat{\pi}_3 = 0.32$,

$\hat{\pi}_4 = 0.15$, $\hat{\pi}_5 = 0.18$. Using the estimated parameters of Model (11) as initial values in the numerical optimization routine, we get the fitted model

$$\sigma_t^2 = \begin{cases} 0.0008 + 0.15 \epsilon_{t-1}^2 + 0.78 \sigma_{t-1}^2 & \text{when } T_t < 6, \\ 0.0001 + 0.00 \epsilon_{t-1}^2 + 0.80 \sigma_{t-1}^2 & \text{when } 6 \leq T_t \leq 9, \\ 0.0015 + 0.46 \epsilon_{t-1}^2 + 0.21 \sigma_{t-1}^2 & \text{when } 9 < T_t \leq 14, \\ 0.0008 + 0.33 \epsilon_{t-1}^2 + 0.62 \sigma_{t-1}^2 & \text{when } 14 < T_t \leq 16, \\ 0.0003 + 0.04 \epsilon_{t-1}^2 + 0.81 \sigma_{t-1}^2 & \text{when } T_t > 16. \end{cases} \quad (12)$$

The effects already noticed for the middle regime (little persistence and strong convexity of the news impact curve) are more pronounced with this five-regime model. A strong coefficient α is also obtained in the third and fourth regimes. Conversely, the volatility in all other regimes does not much depend on the last observation. Again, the model is globally stable in the second order sense.

The next model introduces regimes for extreme low and high temperatures. Letting $s_t = 1$ when $T_t < 3.2$, $s_t = 2$ when $T_t \in [3.2, 9]$, $s_t = 3$ when $T_t \in [9, 14]$, $s_t = 4$ when $T_t \in [14, 18.5]$, and $s_t = 5$ when $T_t > 18.5$, the regime frequencies are given by $\hat{\pi}_1 = 0.06$, $\hat{\pi}_2 = 0.29$, $\hat{\pi}_3 = 0.32$, $\hat{\pi}_4 = 0.28$, $\hat{\pi}_5 = 0.05$. The fitted model is

$$\sigma_t^2 = \begin{cases} 0.0036 + 0.38 \epsilon_{t-1}^2 + 0.48 \sigma_{t-1}^2 & \text{when } T_t < 3.2, \\ 0.0007 + 0.04 \epsilon_{t-1}^2 + 0.68 \sigma_{t-1}^2 & \text{when } 3.2 \leq T_t \leq 9, \\ 0.0004 + 0.30 \epsilon_{t-1}^2 + 0.62 \sigma_{t-1}^2 & \text{when } 9 < T_t \leq 14, \\ 0.0004 + 0.20 \epsilon_{t-1}^2 + 0.72 \sigma_{t-1}^2 & \text{when } 14 < T_t \leq 18.5, \\ 0.0000 + 0.00 \epsilon_{t-1}^2 + 0.92 \sigma_{t-1}^2 & \text{when } T_t > 18.5. \end{cases} \quad (13)$$

Note that the impact of recent observations on the volatility is stronger in the first regime, which reflects the sensitivity of the prices to low temperatures. When temperature becomes extremely low, the consumption for heating utility increases and in such a context the market bid can be saturated. The prices thus can move very quickly, and the volatility locally increases.

The next model allows for 7 regimes, displayed in Eq. (14), with regime frequencies given by $\hat{\pi}_1 = 0.06$, $\hat{\pi}_2 = 0.10$, $\hat{\pi}_3 = 0.19$, $\hat{\pi}_4 = 0.32$, $\hat{\pi}_5 = 0.15$, $\hat{\pi}_6 = 0.13$, $\hat{\pi}_7 = 0.05$. The fitted model is

$$\sigma_t^2 = \begin{cases} 0.0033 + 0.34 \epsilon_{t-1}^2 + 0.52 \sigma_{t-1}^2 & \text{when } T_t < 3.2, \\ 0.0009 + 0.11 \epsilon_{t-1}^2 + 0.65 \sigma_{t-1}^2 & \text{when } 3.2 \leq T_t \leq 6, \\ 0.0001 + 0.00 \epsilon_{t-1}^2 + 0.81 \sigma_{t-1}^2 & \text{when } 6 < T_t \leq 9, \\ 0.0013 + 0.40 \epsilon_{t-1}^2 + 0.30 \sigma_{t-1}^2 & \text{when } 9 < T_t \leq 14, \\ 0.0003 + 0.28 \epsilon_{t-1}^2 + 0.80 \sigma_{t-1}^2 & \text{when } 14 < T_t \leq 16, \\ 0.0006 + 0.60 \epsilon_{t-1}^2 + 0.0003 \sigma_{t-1}^2 & \text{when } 16 < T_t \leq 18.5, \\ 0.0000 + 0.0000 \epsilon_{t-1}^2 + 0.95 \sigma_{t-1}^2 & \text{when } T_t > 18.5. \end{cases} \quad (14)$$

Finally, for the sake of comparison, we estimated a purely periodic four regime model. Noting by Q_i the i -th quarter of the year, and

letting $s_t = i$ when $t \in Q_i$ the regime frequencies are equal to $\pi_i = \frac{1}{4}$ for all $i \in \{1, \dots, 4\}$. The estimated model is

$$\sigma_t^2 = \begin{cases} 0.0001 + 0.20 \epsilon_{t-1}^2 + 0.77 \sigma_{t-1}^2 & \text{when } t \in Q_1 \\ 0.0023 + 0.09 \epsilon_{t-1}^2 + 0.000 \sigma_{t-1}^2 & \text{when } t \in Q_2 \\ 0.000 + 0.01 \epsilon_{t-1}^2 + 0.99 \sigma_{t-1}^2 & \text{when } t \in Q_3 \\ 0.0004 + 0.25 \epsilon_{t-1}^2 + 0.69 \sigma_{t-1}^2 & \text{when } t \in Q_4. \end{cases} \quad (15)$$

This model clearly separates properties of the spring and summer periods ($Q_2 + Q_3$) from the autumn and winter ($Q_1 + Q_4$). The latter periods display greater values of the $\alpha(\cdot)$ coefficients and strong persistence.

5.1. Models comparisons based on tests

The likelihoods of the different models, displayed in Table 4 allow comparing the different fits. From Likelihood Ratio (LR) tests, at the 5% significance level,

- the standard GARCH(1,1) model is not rejected against the 3 regime model;
- the GARCH(1,1) model is however rejected against any model with $d > 3$;
- the 3 regime model is rejected against the 5 regime Models (12) and (13) and 7 regime Model (14).
- the Periodic model is accepted against the GARCH(1,1).

Wald tests not reported here lead to the same conclusions. In the same table, the estimated kurtosis of the variable $\eta_t = \epsilon_t / \sigma_t$ are reported. The biggest kurtosis reduction is obtained for the 7-regime Model (14).

5.2. Models comparisons based on the predictions of the squares

We now study how well the different models perform for out-of-sample prediction of the squares.

5.2.1. Theoretical predictions

Formulas for optimal predictions at horizon $h \geq 1$ can be derived as in standard GARCH models. For the standard GARCH(1,1), the optimal prediction of ϵ_t^2 in the L^2 sense, $E(\epsilon_t^2 | \{\epsilon_{t-\ell}^2, \ell > 0\})$, is obtained from the ARMA(1,1) representation for the squares. Similarly, for Model (3), letting $u_t = \epsilon_t^2 - \sigma_t^2 = (\eta_t^2 - 1)\sigma_t^2$ we have

$$\epsilon_t^2 = \omega(s_t) + (\alpha + \beta)(s_t)\epsilon_{t-1}^2 + u_t - \beta(s_t)u_{t-1}.$$

Letting $\delta_t = \epsilon_t^2 - \omega(s_t) - (\alpha + \beta)(s_t)\epsilon_{t-1}^2$, we thus have³

$$\epsilon_t^2 = \omega(s_t) + (\alpha + \beta)(s_t)\epsilon_{t-1}^2 - \sum_{k=0}^{\infty} \beta(s_t) \dots \beta(s_{t-k}) \delta_{t-k-1} + u_t. \quad (16)$$

³ This representation is valid because (4) implies $\sum_{j=1}^d \pi_j \log \beta(e_j) \leq \sum_{j=1}^d \pi_j E\{\log a(e_j, \eta_0)\} < 0$, from which the existence of the infinite sum in (16) is deduced, by the arguments used to establish the stability condition (see RZ).

Table 4

Log-likelihoods of the estimated models and kurtosis of the standardized returns.

	GARCH ($d=1$)	Model (11) ($d=3$)	Model (12) ($d=5$)	Model (13) ($d=5$)	Model (14) ($d=7$)	Periodic Model (15) ($d=4$)
Log-likelihood	5173	5179	5206	5210	5223	5217
$\hat{\kappa}_\eta$	5.9108	5.7097	5.4021	5.5492	5.2636	5.3777

Note that the expectation of u_t conditional on the past of ϵ_t^2 is zero. The optimal predictor $\hat{\epsilon}_t^2$ of ϵ_t^2 , in the L^2 sense, is then

$$\hat{\epsilon}_t^2 = \omega(s_t) + (\alpha + \beta)(s_t)\epsilon_{t-1}^2 - \sum_{k \geq 0} \beta(s_t) \dots \beta(s_{t-k}) \delta_{t-k-1}.$$

Predictions at higher horizons can be derived similarly. Denote by $P_{t-1}\epsilon_{t+h}^2$, for $h \geq 0$, the optimal predictor of ϵ_{t+h}^2 conditional on the information available at time $t-1$. The $h+1$ -step ahead predictions are related to the 1-step ahead prediction $\hat{\epsilon}_t^2 = P_{t-1}\epsilon_t^2$ by

$$P_{t-1}\epsilon_{t+h}^2 = \omega(s_{t+h}) + \prod_{i=1}^h (\alpha + \beta)(s_{t+i}) \hat{\epsilon}_t^2. \quad (17)$$

Contrary to standard GARCH models, the prediction formulas are time-dependent through the coefficient s_t .

5.2.2. Errors of prediction of the estimated models

The different models were re-estimated over the same sample except the last 500 observations, which were kept for prediction comparisons. The estimated models over the smaller sample turned out to be very close to those estimated on the whole sample. Let n denote the number of observations used for the estimation. To compare the forecasting performances of the previous models, we introduce the criteria, for $i \in \{1, 2, 3\}$ and $h = 1, 40$,

$$m(i, h) = (501-h)^{-1} \sum_{t=n}^{n+500-h} |\epsilon_{t+h}^2 - \hat{P}_{t-1}\epsilon_{t+h}^2|^i$$

where the $\hat{P}_{t-1}\epsilon_{t+h}^2$ are obtained by replacing the coefficients by their estimates in Eq. (17). The Mean-Squared Errors (MSE) of prediction at horizon 1, $m(2, 1)$, are displayed in Table 5 for all fitted models except the three-regime model. It is seen that the standard GARCH(1,1) and the periodic model are outperformed by the models in which the temperature is used, in particular the five- and seven-regime models Eqs. (13) and (14).

Fig. 7 displays the function $m(i, h)$ for $i = 1, 3$, and h ranging from 1 to 40 days. Whatever the criterion used, the 7-regime model is seen to minimize the prediction errors, for short and long horizons. While the standard GARCH provides satisfactory results, the periodic GARCH model is clearly beaten for long term predictions.

5.3. Conditional and unconditional variance

In standard GARCH models it is important to distinguish the conditional variance (or volatility) σ_t^2 from the unconditional one which, when existing, is given by $E\epsilon_t^2 = \omega/(1 - \alpha - \beta)$.

By contrast, for the model of this paper, the volatility is still σ_t^2 but the unconditional variance, as a function of the s_{t-i} , $i \geq 0$, is no longer

a constant. It can be recursively obtained, provided Eq. (5) is satisfied, from

$$E\epsilon_t^2 = \omega(s_t) + \mu(s_t)E\epsilon_{t-1}^2 \quad (18)$$

where $\mu(s_t) = \alpha(s_t) + \beta(s_t)$. For Models (9), (13), (14) and (15), Fig. 8 displays the volatilities, while Fig. 9 displays the trajectories of $E\epsilon_t^2$ obtained from Eq. (18) after initialization.

We emphasize the fact that those time-varying moments are not volatilities: they are not conditional to the past price variations, their fluctuations being entirely determined by the temperature variations. It is seen that those fluctuations can be huge, in particular for the 5-regime Model (13).

5.4. Conditional and unconditional kurtosis

For standard GARCH models, the *conditional* kurtosis coefficient, defined as the ratio of the fourth-order conditional moment on the squared conditional variance, is constant and is equal to the kurtosis κ_η of the independent process (η_t) . The *unconditional* kurtosis coefficient, when existing, is equal to κ_η multiplied by a constant greater than 1, and can be used to measure the leptokurticity of the unconditional distribution.

For the model of this paper, it is interesting to compare the conditional and unconditional kurtosis coefficients with those obtained for the standard GARCH. The fourth unconditional moments of ϵ_t can be computed recursively, for $t \geq 1$, using Eq. (18) and

$$E\epsilon_t^4 = \omega^2(s_t)\kappa_\eta + \lambda(s_t)E\epsilon_{t-1}^4 + 2\omega(s_t)\mu(s_t)E\epsilon_{t-1}^2\kappa_\eta,$$

with initial values $E\epsilon_0^2$ and $E\epsilon_0^4$. It follows that the unconditional kurtosis of ϵ_t , defined as $\kappa_t = E\epsilon_t^4/(E\epsilon_t^2)^2$, satisfies the recursion

$$\kappa_t - \kappa_\eta = \left(\frac{E\epsilon_{t-1}^2}{E\epsilon_t^2} \right)^2 \left\{ (\kappa_{t-1} - \kappa_\eta)\lambda(s_t) + \kappa_\eta(\kappa_\eta - 1)\alpha^2(s_t) \right\} \quad (19)$$

and does not converge to a constant as t tends to infinity. On the contrary, the conditional kurtosis of ϵ_t is simply given by

$$\frac{E(\epsilon_t^4 | \epsilon_s, s < t)}{\{E(\epsilon_t^2 | \epsilon_s, s < t)\}^2} = \frac{\sigma_t^4 E\eta_t^4}{(\sigma_t^2 E\eta_t^2)^2} = \kappa_\eta,$$

as in the standard GARCH(1,1) case.

These computations allow obtaining the time-varying unconditional fourth-order moments, provided that they exist. Recall that, in view of Eq. (5), the existence condition is $\prod_{j=1}^d \lambda(e_j)^{\eta_j} < 1$. In view of Table 6, this condition is satisfied for the 5 and 7-regime Models (13) and (14), and the periodic Model (15), if we assume that $\kappa_\eta = 3$. Fig. 10 shows the unconditional kurtosis in logarithms, recursively computed using Eq. (19). The fluctuations are huge and the graph allows visualizing the fact that the unconditional distribution fluctuates with temperature.

Table 5MSE ($\times 10^{-5}$) of predictions at horizon 1 (last 500 observations).

GARCH ($d=1$)	Model (12) ($d=5$)	Model (13) ($d=5$)	Model (14) ($d=7$)	Periodic Model (15) ($d=4$)
9.319	9.206	9.014	9.051	9.259

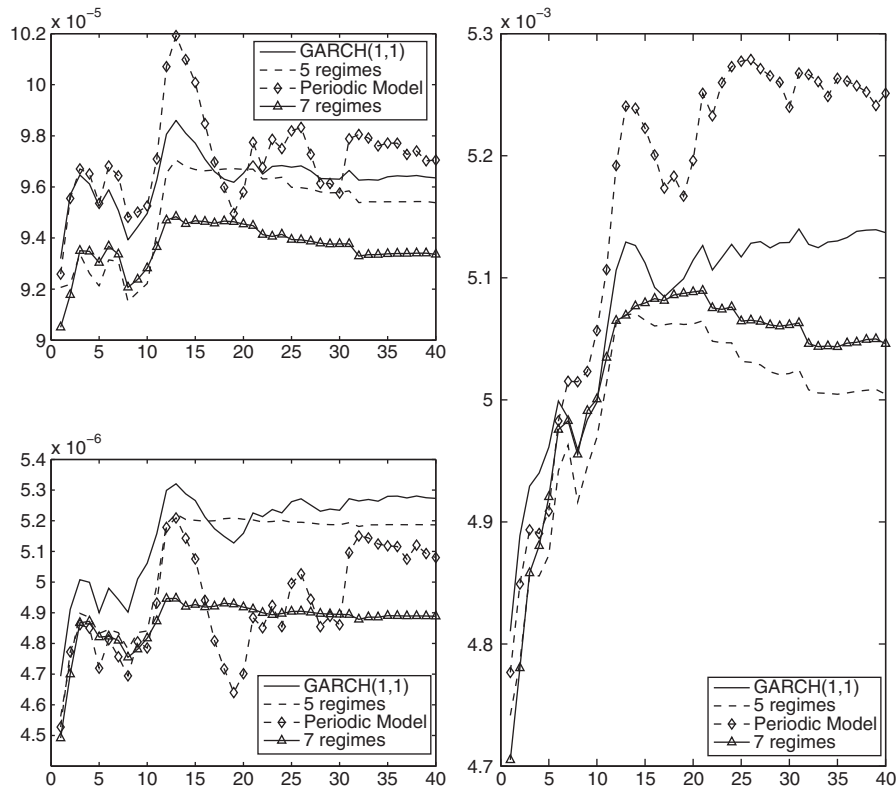


Fig. 7. Comparisons of prediction errors at horizon $h = 1, \dots, 40$. Mean squared-errors, $m(2, h)$ (top left panel), absolute errors to the power 3, $m(3, h)$ (bottom left panel) and absolute errors, $m(1, h)$ (right panel).

5.5. Including asymmetric effects

The model of this paper is amenable to extensions allowing for asymmetries in volatility. For financial data, it has been extensively documented that past negative returns have a bigger impact on the current volatility than past positive returns of the same magnitude. Several formulations have been proposed to capture this effect, in particular the Exponential GARCH (EGARCH) model of Nelson (1991) and the Threshold GARCH (TGARCH) model of Glosten, Jaganathan

and Runkle (1993) and Zakoïan (1994). Estimating a three-regime model of this type, with the same temperature regimes as in Eq. (11), yields

$$\sigma_t^2 = \begin{cases} 0.0003 + \frac{0.23}{(0.10)} (\epsilon_{t-1}^+)^2 + \frac{0.00}{(0.04)} (\epsilon_{t-1}^-)^2 + \frac{0.82}{(0.07)} \sigma_{t-1}^2 & \text{when } T_t < 9, \\ 0.0002 + \frac{0.19}{(0.08)} (\epsilon_{t-1}^+)^2 + \frac{0.24}{(0.09)} (\epsilon_{t-1}^-)^2 + \frac{0.75}{(0.084)} \sigma_{t-1}^2 & \text{when } 9 \leq T_t \leq 14, \\ 0.0003 + \frac{0.00}{(0.03)} (\epsilon_{t-1}^+)^2 + \frac{0.38}{(0.11)} (\epsilon_{t-1}^-)^2 + \frac{0.76}{(0.07)} \sigma_{t-1}^2 & \text{when } T_t > 14. \end{cases}$$

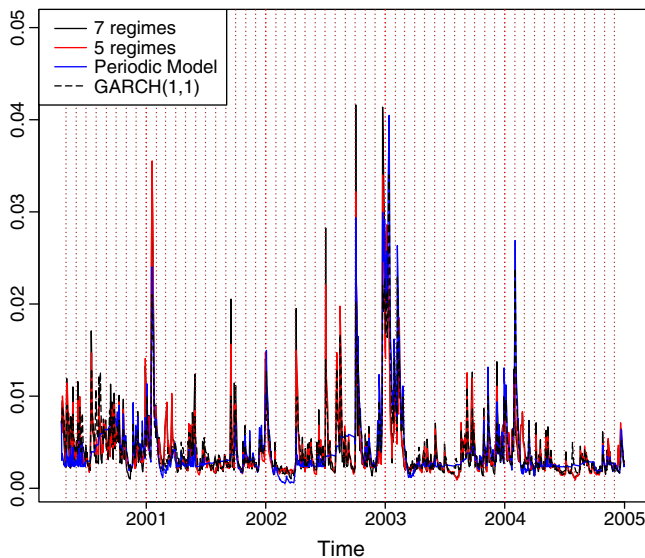


Fig. 8. Volatility (σ_t^2) for Models (9), (13), (14) and (15).

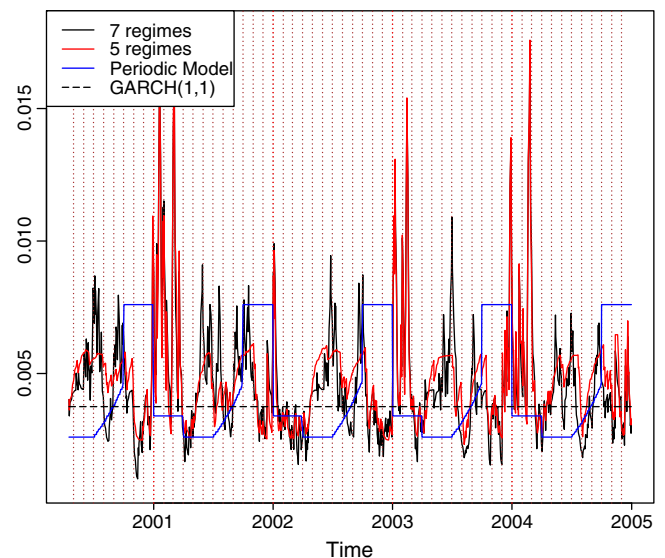


Fig. 9. Unconditional variance $E\epsilon_t^2$ for Models (9), (13), (14) and (15).

Table 6
Coefficient $\prod_{j=1}^d \lambda(\epsilon_j)^{\eta_j}$ involved in the existence of fourth-order moments.

GARCH	Model (12)	Model (13)	Model (14)	Periodic Model (15)
1.0003	1.0285	0.9286	0.9547	0.5071

where $x^+ = \max(x, 0)$, $x^- = \min(x, 0)$. The estimated coefficients of $(\epsilon_{t-1}^+)^2$ and $(\epsilon_{t-1}^-)^2$ in regimes 1 and 3 clearly indicate that asymmetry is present in the data, though in opposite directions for the two regimes. Asymmetric extensions of Models (12)–(15) were estimated. For brevity we only report the log likelihoods in Table 7. Comparison with Table 4 shows that, for given temperature regimes, asymmetric models are always preferred to the symmetric ones. Among the asymmetric formulations, the model with $d=7$ regimes passes the LR tests against all smaller models and thus seems to be the most satisfactory.

6. Conclusion

This paper reviewed a class of GARCH models allowing volatility to depend on an observed exogenous process. This observability of the state variable makes the model much easier to use than the so-called Markov-switching processes, in which the regime change is governed by a latent Markov chain. The model can be estimated by QML and a consistent estimator of the asymptotic covariance matrix has been proposed. The methodology has been applied to the residuals of the gas price equation in a VECM, using the temperature as exogenous variable. We found evidence of five regimes, with a very different volatility dynamics in the moderate-temperature regime. The model can be used for prediction purposes, using temperature scenarios. Asymmetric extensions allowing for different impacts of past positive and negative values were found fruitful for the volatility of gas price residuals. Other extensions, by including more lags in the volatility or by considering multivariate series, could be considered. Another topic of interest concerns the estimation of the regimes. These extensions are left for future research. It is hoped that the article will broaden the use of time series models driven by exogenous variables.

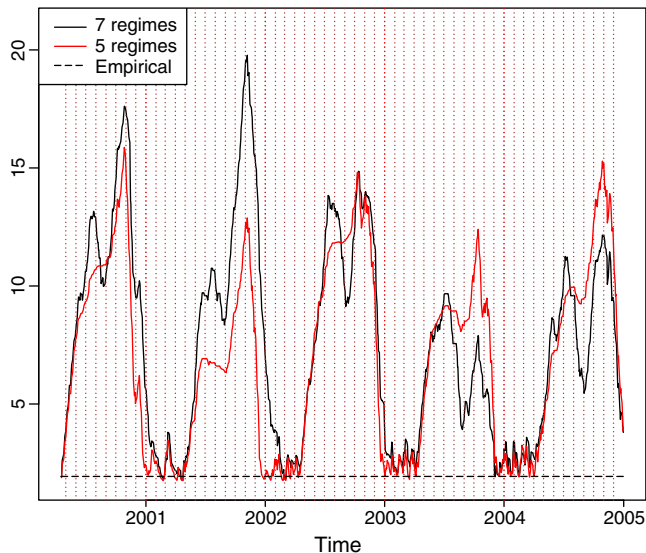


Fig. 10. Unconditional time-varying kurtosis in logarithms, $\log\{E\epsilon_t^4/E(\epsilon_t^2)^2\}$, for Models (13) and (14). The dotted line corresponds to the empirical value, computed over the whole sample, obtained by replacing expectations by empirical moments.

Table 7
Log-likelihoods of the estimated models with asymmetric effects.

Asymmetric extensions of	GARCH ($d=1$)	(11) ($d=3$)	(12) ($d=5$)	(13) ($d=5$)	(14) ($d=7$)	(15) ($d=4$)
Log-likelihood	5176	5239	5260	5260	5274	5254

Appendix A. Technical details

Proof of Proposition 1. For all $\theta \in \Theta$, let

$$\tilde{J}_n(\theta) = \frac{1}{n} \sum_{t=1}^n \frac{1}{\tilde{\sigma}_t^4(\theta)} \frac{\partial \tilde{\sigma}_t^2(\theta)}{\partial \theta} \frac{\partial \tilde{\sigma}_t^2(\theta)}{\partial \theta'}, \quad J_n(\theta) = \frac{1}{n} \sum_{t=1}^n \frac{1}{\sigma_t^4(\theta)} \frac{\partial \sigma_t^2(\theta)}{\partial \theta} \frac{\partial \sigma_t^2(\theta)}{\partial \theta'}.$$

Note that $\hat{J}_n = \tilde{J}_n(\hat{\theta}_n)$. We have, letting $\theta = (\theta_i)_{i=1, \dots, 3d}$,

$$\begin{aligned} & \frac{1}{n} \sum_{t=1}^n \left\{ \frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta_i} \frac{\partial \sigma_t^2}{\partial \theta_j} \right\}_{\theta=\hat{\theta}_n} \\ &= \frac{1}{n} \sum_{t=1}^n \left\{ \frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta_i} \frac{\partial \sigma_t^2}{\partial \theta_j} \right\}_{\theta=\theta_0} + \frac{1}{n} \sum_{t=1}^n \left\{ \frac{\partial}{\partial \theta'} \left(\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta_i} \frac{\partial \sigma_t^2}{\partial \theta_j} \right) \right\}_{\theta=\theta_{ij}} (\hat{\theta}_n - \theta_0). \end{aligned} \quad (\text{A.1})$$

where θ_{ij} is between $\hat{\theta}_n$ and θ_0 . Denote by $(\sigma_{s,t}^2(\theta))$ the process recursively defined by $\sigma_{s,t}^2(\theta) = \omega(S_t) + \alpha(S_t)\epsilon_{t-1}^2 + \beta(S_t)\sigma_{s,t-1}^2(\theta)$. We have, for almost all sequence (s_t) ,

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \left\| n^{-1} \sum_{t=1}^n \frac{\partial}{\partial \theta'} \left(\frac{1}{\sigma_t^4(\theta_{ij}^*)} \frac{\partial \sigma_t^2(\theta_{ij}^*)}{\partial \theta_i} \frac{\partial \sigma_t^2(\theta_{ij}^*)}{\partial \theta_j} \right) \right\| \\ & \leq \limsup_{n \rightarrow \infty} n^{-1} \sum_{t=1}^n \sup_{\theta \in \mathcal{V}(\theta_0)} \left\| \frac{\partial}{\partial \theta'} \left(\frac{1}{\sigma_t^4(\theta)} \frac{\partial \sigma_t^2(\theta)}{\partial \theta_i} \frac{\partial \sigma_t^2(\theta)}{\partial \theta_j} \right) \right\| \\ & = E_{\theta_0} \sup_{\theta \in \mathcal{V}(\theta_0)} \left\| \frac{\partial}{\partial \theta'} \left(\frac{1}{\sigma_{s,t}^4(\theta)} \frac{\partial \sigma_{s,t}^2(\theta)}{\partial \theta_i} \frac{\partial \sigma_{s,t}^2(\theta)}{\partial \theta_j} \right) \right\| < \infty. \end{aligned}$$

where $\|\cdot\|$ denotes any norm on $\mathbb{R}^{3d}$. The equality follows from Lemma 5.2 in RZ and the fact that $\sigma_t^2(\theta)$ and $\frac{\partial \sigma_t^2(\theta)}{\partial \theta}$ are measurable functions of $(s_t, s_{t-1}, \dots, \eta_t, \eta_{t-1}, \dots)$. The last inequality is a consequence of iii), in the proof of Theorem 4.2 in RZ. Since $\hat{\theta}_n - \theta_0 \rightarrow 0$ a.s., the last term in (20) converges to zero in probability as n tends to infinity.

Using again Lemma 5.2 in RZ, we obtain the a.s. convergence to J of the first term in the right-hand side of (20). Thus we have shown that

$$\frac{1}{n} \sum_{t=1}^n \left\{ \frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta_i} \frac{\partial \sigma_t^2}{\partial \theta_j} \right\}_{\theta=\hat{\theta}_n} \rightarrow J, \quad \text{a.s.}$$

Since, by FZ, Proof of Theorem 4.2,

$$\sup_{\theta \in \mathcal{V}(\theta_0)} \left\| \frac{1}{n} \sum_{t=1}^n \left\{ \frac{\partial^2 \tilde{\ell}_t(\theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 \ell_t(\theta)}{\partial \theta \partial \theta'} \right\} \right\| \rightarrow 0 \quad \text{a.s.}$$

where $\ell_t(\theta)$ is defined as $\tilde{\ell}_t(\theta)$ with $\tilde{\sigma}_t$ replaced by σ_t , we thus have

$$\hat{J}_n = \frac{1}{n} \sum_{t=1}^n \left\{ \frac{1}{\tilde{\sigma}_t^4} \frac{\partial \tilde{\sigma}_t^2}{\partial \theta_i} \frac{\partial \tilde{\sigma}_t^2}{\partial \theta_j} \right\}_{\theta=\hat{\theta}_n} \rightarrow J, \quad \text{a.s.}$$

By the same arguments we prove that

$$\hat{\kappa}_\eta = \frac{1}{n} \sum_{t=1}^n \frac{\epsilon_t^4}{\tilde{\sigma}_t^4(\hat{\theta}_n)} \rightarrow E\eta_t^4$$

and the proposition is established.

Appendix B. An Asymmetric CCC GARCH

Let $X_t = (\epsilon_t, \zeta_t, \xi_t)$. We consider a CCC Asymmetric GARCH(1, 1) model defined by

$$\begin{cases} X_t &= H_t^{1/2} \eta_t \\ H_t &= D_t R_0 D_t, \quad D_t = \text{diag}(\sqrt{h_{1,t}}, \sqrt{h_{2,t}}, \sqrt{h_{3,t}}), \\ \underline{h}_t &= (h_{1,t}, h_{2,t}, h_{3,t})' = \underline{\omega}_0 + A_{0,+} X_{t-1}^+ + A_{0,-} X_{t-1}^- + B_0 \underline{h}_{t-1}, \end{cases} \quad (21)$$

where $X_t^+ = (\{\epsilon_t^+\}^2, \{\zeta_t^+\}^2, \{\xi_t^+\}^2)$, $X_t^- = (\{\epsilon_t^-\}^2, \{\zeta_t^-\}^2, \{\xi_t^-\}^2)$ and $x^+ = \max(x, 0) = (-x)^-$, R_0 is a correlation matrix, $A_{0,+}$, $A_{0,-}$ and B_0 are 3×3 matrices with positive coefficients, and (η_t) is an iid sequence of centered variables on $\mathbb{R}^3$ with identity covariance matrix. Estimation of the model by QML (see Francq and Zakoian (forthcoming)) yields the following estimates:

$$\underline{\omega}_0 = \begin{pmatrix} 0.0004 \\ 0.0000 \\ 0.0625 \end{pmatrix}, \quad A_{0,+} = \begin{pmatrix} 0.15 & 0.00 & 0.00 \\ 0.00 & 0.03 & 0.00 \\ 0.00 & 0.00 & 0.00 \end{pmatrix}, \quad A_{0,-} = \begin{pmatrix} 0.20 & 0.00 & 0.00 \\ 0.00 & 0.09 & 0.00 \\ 0.00 & 0.00 & 0.06 \end{pmatrix}$$

$$B_0 = \begin{pmatrix} 0.74 & 0.00 & 0.00 \\ 0.00 & 0.89 & 0.00 \\ 0.00 & 0.00 & 0.95 \end{pmatrix}, \quad R_0 = \begin{pmatrix} 1.0000 & 0.0024 & -0.0941 \\ 0.0024 & 1.0000 & 0.0230 \\ -0.0941 & 0.0230 & 1.0000 \end{pmatrix}.$$

References

- Aknouche, A., Bibi, B., 2009. Quasi-maximum likelihood estimation of periodic GARCH and periodic ARMA-GARCH processes. *Journal of Time Series Analysis* 30, 19–46.
- Aknouche, A., Rabehi, N., 2010. On an independent and identically distributed mixture bilinear time series model. *Journal of Time Series Analysis* 31, 113–131.
- Asche, F., Osmundsen, P., Sandsmark, M., 2006. The UK market for natural gas, oil and electricity: are the prices decoupled? *Energy Journal* 27, 27–40.
- Azrak, R., Mèlard, G., 2006. Asymptotic properties of quasi-maximum likelihood estimators for ARMA models with time-dependent coefficients. *Statistical Inference for Stochastic Processes* 3, 279–330.
- Bachmeier, L.J., Griffin, J.M., 2006. Testing for market integration : crude oil, coal, and natural gas. *The Energy Journal* 27, 55–71.
- Berkes, I., Horváth, L., Kokoszka, P., 2003. GARCH processes: structure and estimation. *Bernoulli* 9, 201–227.
- Bibi, A., Francq, C., 2003. Consistent and asymptotically normal estimators for time-dependent linear models. *Annals of the Institute of Statistical Mathematics* 55, 41–68.
- Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* 31, 307–327.
- Bollerslev, T., Ghysels, E., 1996. Periodic autoregressive conditional heteroskedasticity. *Journal of Business and Economic Statistics* 14, 139–151.
- Elliott, G., Rothenberg, T.J., Stock, J.H., 1996. Efficient tests for an autoregressive unit root. *Econometrica* 64, 813–836.
- Engle, R.F., 1982. Autoregressive conditional heteroskedasticity with estimates of the variance of the United Kingdom inflation. *Econometrica* 50, 987–1007.
- Francq, C., Gautier, A., 2004a. Large sample properties of parameter least squares estimates for time-varying ARMA models. *Journal of Time Series Analysis* 25, 765–783.
- Francq, C., Gautier, A., 2004b. Estimation of time-varying ARMA models with Markovian changes in regime. *Statistics and Probability Letters* 70, 243–251.
- Francq, C., Zakoian, J.-M., 2004. Maximum likelihood estimation of pure GARCH and ARMA-GARCH processes. *Bernoulli* 10, 605–637.
- Francq, C., Zakoian, J.-M., 2009. Bartlett's formula for a general class of non linear processes. *Journal of Time Series Analysis* 30, 449–465.
- Francq, C. and Zakoian, J.-M. 2011. QML Estimation of a Class of Multivariate Asymmetric Garch Models. Forthcoming in *Econometric Theory*.
- Frühwirth-Schnatter, S., 2006. *Finite Mixture and Markov Switching Models*. Springer Verlag.
- Glosten, L.R., Jagannathan, R., Runkle, D., 1993. On the relation between the expected values and the volatility of the nominal excess return on stocks. *The Journal of Finance* 48, 1779–1801.
- Hamilton, J.D., 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 57, 357–384.
- Hamilton, J.D., Susmel, R., 1994. Autoregressive conditional heteroskedasticity and changes in regime. *Journal of Econometrics* 64, 307–333.
- Hylleberg, S., Engle, R.F., Granger, C.W.J., Yoo, B.S., 1990. Seasonal integration and cointegration. *Journal of Econometrics* 44, 215–238.
- Johansen, S., 1988. Statistical analysis of cointegration vectors. *Journal of Economic Dynamics and Control* 12, 231–254.
- Johansen, S., 1995. *Likelihood-based Inference in Cointegrated Vector Autoregressive Models*. Oxford University Press, Oxford.
- Koopman, S.J., Ooms, M., Carnero, M.A., 2007. Periodic seasonal Reg-ARFIMA-GARCH models for daily electricity spot prices. *Journal of the American Statistical Association* 102, 16–27.
- Kwiatkowski, D., Phillips, P.C.B., Schmidt, P., Shin, Y., 1992. Testing the null hypothesis of stationarity against the alternative of a unit root: how sure are we that economic time series have a unit root? *Journal of Econometrics* 54, 159–178.
- Li, C.W., Li, W.K., 1996. On a double-threshold autoregressive heteroscedastic time series model. *Journal of Applied Econometrics* 11, 253–274.
- Nelson, D.B., 1991. Conditional heteroskedasticity in asset returns: a new approach. *Econometrica* 59, 347–370.
- Osborn, D.R., Savva, C.S., Gill, L., 2008. Periodic dynamic conditional correlations between stock markets in Europe and the US. *Journal of Financial Econometrics* 6, 307–325.
- Panagiotidis, T., Rutledge, E., 2007. Oil and gas markets in the UK: evidence from a cointegrating approach. *Energy Economics* 29, 329–347.
- Regnard, N., Zakoian, J.-M., 2010. Structure and estimation of a class of nonstationary yet nonexplosive GARCH models. *Journal of Time Series Analysis* 31, 348–364.
- Said, S.E., Dickey, D., 1984. Testing for unit roots in autoregressive moving-average models with unknown order. *Biometrika* 71, 599–607.
- Susmel, R., Thompson, A., 1997. Volatility, storage and convenience: evidence from natural gas markets. *Journal of Futures Markets* 17, 17–43.
- Zakoian, J.-M., 1994. Threshold heteroskedastic models. *Journal of Economic Dynamics and Control* 18, 931–955.